Policy and Artificial Intelligence

Analyzing public comments on the Federal Trade Commission's 2026 personalized-pricing proposal

Ruth & Ketty · INFO 4340 Final Project

Our overall question: Can generative artificial intelligence help a public agency make sense of thousands of public comments accurately, affordably, and responsibly?

1. The policy problem shaped the measurement problem

In August 2026, the Federal Trade Commission proposed an enforcement policy statement on personalized pricing - the use of personal data to set prices according to what a firm believes an individual consumer is willing to pay. The Commission did not propose a blanket ban. It stated that Congress had not granted authority to prohibit personalized pricing in all circumstances, while explaining that undisclosed or misleading personalization could violate laws against unfair or deceptive practices. The proposal emphasized clear and conspicuous disclosure of the fact that a price is personalized, the basis of personalization, and the types of data used.[1]

That distinction gave us the central analytical problem. A negative comment about personalized pricing does not necessarily tell us what the commenter wants the government to do. One person may want disclosure, another may want a ban, another may accept personalized discounts but oppose individualized markups, and another may oppose government intervention. We therefore measured policy intent rather than generic sentiment. Our specific classification question became: "What response does each commenter want from the Federal Trade Commission?"

2. We built human judgment before automated judgment

Our analytic corpus contains 1,566 individual, text-based comments from docket FTC-2026-1057. We treat it as a verified analytic subset, not the entire docket, and we do not interpret submission frequencies as representative public opinion because public comments are self-selected.

We first used a 40-comment development pilot to expose ambiguous boundaries in our coding dictionary. After revising and freezing the rules, we independently coded the same 120-comment random validation sample without seeing each other's labels. We agreed on 102 of 120 stance labels (85.0%), and Cohen's kappa was 0.706, above the course threshold of 0.60. Only then did we reconcile disagreements and create the gold-standard labels used to evaluate automated systems.

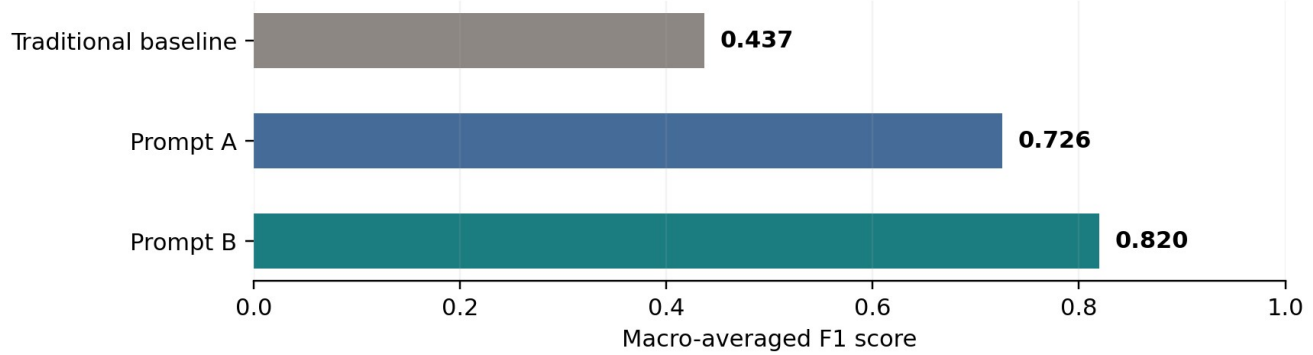
85.0%	0.706	120
independent stance agreement	Cohen's kappa	gold-standard validation comments

The largest disagreement boundary was between "unclear" and "stronger action." We froze a conservative rule: saying that personalized pricing is unfair or wrong is not automatically a request for government action. Direct requests to ban, stop, protect consumers, guarantee an opt-out, or otherwise strengthen the remedy count as stronger action. This human coding work later became the logic encoded in Prompt B.

3. We compared three automated approaches

We did not evaluate a generative model in isolation. First, we built a traditional text-classification baseline using term frequency-inverse document frequency features and class-weighted logistic regression. Its out-of-fold accuracy was 65.8%, and its macro-averaged F1 score was 0.437. The rare mixed-or-conditional category was particularly difficult.

We then evaluated two structured generative-artificial-intelligence prompts on exactly the same 120 reconciled comments. Prompt A supplied a concise taxonomy with minimal guidance. Prompt B translated the frozen human codebook into machine-readable instructions, adding policy context, boundary rules, tie-breakers, and a substantive-reason test for concern labels. During development, we also used Arena.ai for exploratory side-by-side comparison of model responses. We treated Arena.ai as a learning and comparison tool, not as formal performance evidence or as the basis for choosing the final production model.



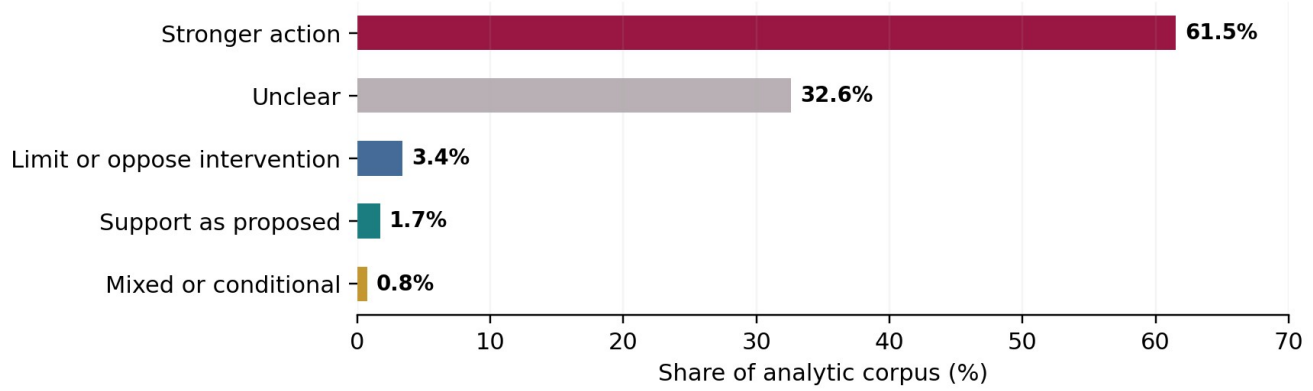
Prompt A had slightly higher raw accuracy (81.7% versus 80.8%), but Prompt B produced much stronger class-balanced stance performance: a macro-averaged F1 score of 0.820 versus 0.726. Prompt B also identified two of the three mixed-or-conditional validation cases, compared with one of three for Prompt A. We selected Prompt B because the project required dependable performance across classes rather than simply maximizing the majority-class accuracy.

The choice was operational rather than a claim of statistical dominance. A paired stratified bootstrap showed that the 95% interval for the Prompt-B-minus-Prompt-A macro-averaged F1 difference crossed zero. Prompt A was also somewhat stronger on multi-label concern extraction, so we interpret concern-level findings more cautiously.

Prompt B also gave us a useful safety valve: it captured 78.3% of its stance errors with the review flag, and accuracy among validation comments it did not flag was 93.5%.

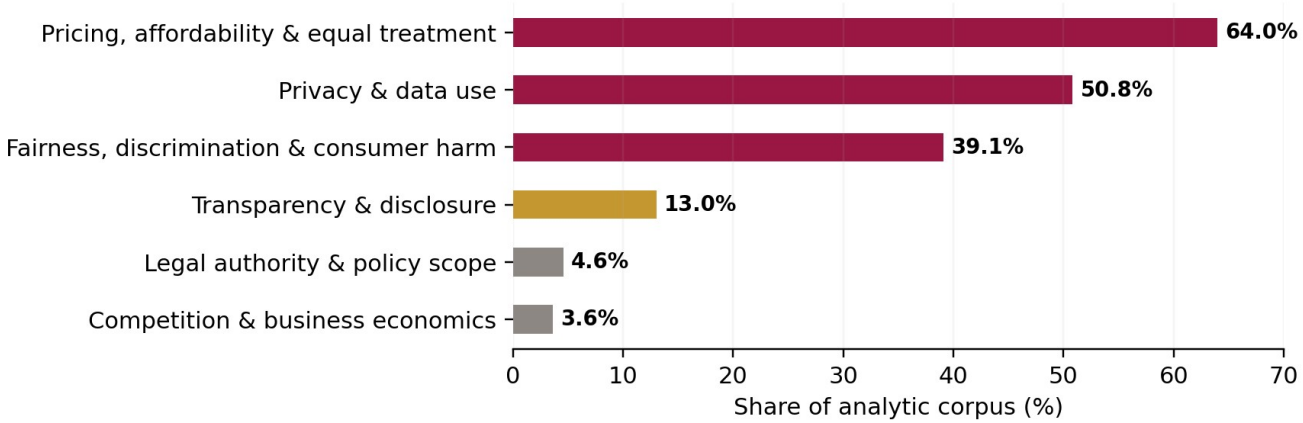
4. What the 1,566 comments showed

After selecting Prompt B, we classified the entire 1,566-comment analytic corpus. The dominant model-coded category was stronger action: 963 comments (61.5%). Another 511 comments (32.6%) were coded unclear. Only 27 comments (1.7%) were classified as supporting the proposal as proposed. These are model-coded frequencies within a self-selected corpus, not population estimates.



5. The substantive concern was broader than disclosure

The concern labels help explain why a large share of comments were classified as asking for stronger action. Pricing, affordability, and equal treatment appeared in 64.0% of comments; privacy and data use in 50.8%; and fairness, discrimination, and consumer harm in 39.1%. Transparency and disclosure appeared in 13.0%. Because concern coding is multi-label, these percentages overlap.



We do not read the 13.0% figure as evidence that disclosure is unimportant. Rather, it shows that the public comments frequently framed the problem in substantive terms - who pays what price, how personal data are used, and whether individualized treatment is fair - rather than as a disclosure problem alone. The coding dictionary also preserved mixed positions, including comments that accepted personalized discounts while opposing individualized markups.

6. The model cost was tiny; responsible analysis was not

The recorded Claude Haiku 5.5 model-inference cost for the project was approximately $0.69: about $0.035 for Prompt A validation, $0.048 for Prompt B validation, $0.581 for the 1,446 new full-corpus calls, and $0.026 for two additional stability runs. At the model level, this shows how inexpensive first-pass classification can be. It does not mean that a public-sector project costs 69 cents. The total cost of responsible use also includes staff time, data engineering, quality assurance, security, documentation, governance, and human review.[2]

> **The value proposition is not "replace the analyst." It is "use inexpensive automation to concentrate analyst attention on the difficult cases."**

7. Governance changed the deployment decision

Our course governance workshop asked us to score risks and convert the most important ones into operating rules that an auditor could actually enforce. We applied that logic directly. Automated aggregate reporting is prohibited if the macro-averaged F1 score falls below 0.80. Human-human Cohen's kappa below 0.60 reopens the codebook. A text-length error-rate ratio above 1.25 times the best-performing group triggers investigation. Invalid evidence quotations are rejected, retried, or sent to human review. Exact three-run stance stability below 90% also triggers investigation.

Prompt B passed the performance floor at 0.820, but the final stability test reached 86.7%, below our 90% target, and the text-length disparity rule was also triggered. We therefore do not authorize autonomous reporting. The model can support aggregate analysis only with human review, monitoring, documentation, and escalation.

The final model/evidence governance queue contains 510 comments (32.6%). This is not the same set as the 511 comments classified unclear. For aggregate research, unclear is a valid category and should not be forced into another stance. For case-level interpretation, an unclear record still requires human judgment.

8. How the ideas connect

The project's main lesson is that the technical model was only one part of the analysis. The policy document told us why ordinary sentiment would be misleading. That measurement choice forced us to define stance categories. Human disagreement then showed us where those categories were ambiguous. Reconciliation converted those lessons into a frozen codebook. Prompt B performed well because it encoded that human-developed boundary logic. Finally, the evaluation results - especially the review flag, stability test, and subgroup differences - determined the governance rules. In other words, the project moved from policy question → measurement → human coding → model comparison → evaluation → governance, rather than beginning with a model and searching for a use.

We also learned that apparent model quality can be deceptive. Arena.ai comparisons were useful for seeing how models differed in wording and reasoning, but a persuasive response is not the same as a validated classifier. The formal benchmark came from our independent human coding. Similarly, low inference cost did not justify autonomous use: our model was inexpensive and useful while still triggering stability and subgroup-monitoring rules.

Our final conclusion: generative artificial intelligence can make large public-comment records more tractable, but only when the coding logic, evaluation criteria, and operating rules are explicit.

Limitations

- The public-comment corpus is self-selected and should not be treated as representative public opinion.
- The 1,566-comment analytic corpus is a verified subset rather than the complete docket; organizational and attachment-based filings are outside the primary classifier.
- The random validation sample contained only three of the five stance categories, so class-balanced validation remains uncertain for the two rare categories that did not appear.
- Concern extraction performed less strongly than stance classification and should be interpreted with greater caution.
- Model prices and model behavior can change, so future use requires revalidation rather than assuming these results transfer unchanged.

Sources

1. Federal Trade Commission. Federal Trade Commission's Proposed Enforcement Policy Statement Regarding Personalized Pricing. August 19, 2026; and related Federal Trade Commission public-comment notice.
2. Anthropic. Claude Haiku 5.5 pricing documentation, checked November 11, 2026. Short prompts up to 100,000 tokens were listed at $0.10 per million input tokens and $0.50 per million output tokens.
3. INFO 4340 Class 17 Governance Checklist Workshop. Risk scoring by likelihood and impact; measurable operating-rule pattern.